Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/04/2004
PI Name:	Pollack, Lois Ph.D.		
Project Title:	Microscale Mixer for Protein Folding		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Batt, Carl (Cornell University)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the un iverse obscured by Earth"s gravity		

The broad, long-term objective of this work is to understand the interactions involved in macromolecular folding and function. It is necessary to trigger and monitor structural changes on time scales as short as microseconds to observe these interesting dynamics. At present, our understanding is limited by the lack of suitable experimental tools accessing dynamics on time scales ranging from microseconds to seconds. In our experiments, we employ a microscale fast fluid mixer with two different probes of macromolecular structure: small angle x-ray scattering to determine global size and shape changes as well as high sensitivity fluoresence microscopy to probe more local interactions and dynamics. The flow cells are compact, and may be adaptable to a gravity-free environment. In previous versions of these flow cells, the macromolecules were in direct contact with the walls defining the micro-flow channels. As a result, some aggregation of protein occured on the surfaces, causing a small, but persistent, background increase as a function of time. In our proposed flow cell design, a layer of buffer solution flows along all of the surface and prevents their

Task Description:	contact with the folding macromolecules. We are also exploring the use of different materials for the cells and windows. Once complete, these new flow cells will permit higher quality measurements by decreasing (or eliminating) extraneous sources of scatter. With this improved signal to background ratio, better quality data can be acquired from a smaller quantity of sample, a critical consideration when dealing with the planned study of mutants. The objective of this project is to study global structural changes that occur during the initial, rapid steps of macromolecular folding. The technique that we have developed monitors rapidly varying molecular conformations by synchrotron x-ray scattering. Sharp time resolution is achieved by the use of a microfabricated flow cell. This technology will be applied to study the folding of both wild-type and variant macromolecules. The latter will be carefully selected to elucidate the role of specific residues in the formation of native, compact structures. Fluorescence microscopy will also be used to monitor local dynamics. A long term goal is the application of both techniques to the same molecule, to acquire both local and global structural information. Our understanding of the dynamics of conformational change in macromolecules is limited by a lack of suitable, real-time experimental probes of their most basic parameters: size and shape. The goal of this project is to probe the dramatic changes that occur at the very initiation of the folding process, specifically the collapse of the flexible, denatured chain into a compact structure. In addition, we have initiated a series of measurements of conformational dynamics of folded proteins on the microsecond time scale. These studies will permit more detailed understanding of the kinetics of physiologically important enzymatic reactions and are outside of the capabilities of commercially available systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This past year, using the vastly improved mixing devices described within the original proposed for this project, we continued to probe the earliest stages of RNA folding. Our previous experiments found two rapid structural changes in the RNA that occured within a fraction of a second after the initiation of folding. This past year, we have identified the physical nature of the most rapid folding phase. It is an electrostatic relaxation of the molecule. To make this determination, we probed the folding of variants, that is RNA molecules that have been altered so that native contacts cannot form. The first phase of the folding of these variants was identical to the first phase of folding of the wildtype molecule. Interestingly, the second phase observed in the wildtype molecule, was absent in the variant, suggeting that intra-molecular contacts form from within an electrostatically relaxed, compact configuration. We have also continued work on a project using anomalous small angle x-ray scattering to determine the spatial distribution of ions around short DNA strands.
Bibliography Type:	Description: (Last Updated: 07/19/2007)
Articles in Peer-reviewed Journals	Das, R., Mills, T.T. Kwok, L.W., Maskel, G.S., Millett, I.S., Doniach, S., Finkelstein, K.D., Herschlag, D., Pollack, L. "Counterion Distribution around DNA Probed by Solution X-ray Scattering " Physical Review Letters , May-2003
Articles in Peer-reviewed Journals	Das, R., Kwok, L.W., Millett, I.S., Bai, Y., Mills, T.T., Jacob, J., Maskel, G.S. , Seifert, S., Mochrie, S.G.J., Thiyagarajan, P, Doniach, S., Pollack, L., Herschlag, D. "The fastest global events in RNA folding: electrostatic relaxation and tertiary collapse of the tetrahymena ribozyme" Journal of Molecular Biology , Jul-2004